

Activity 1

Question: Create a table named StudentDetails. It should have two columns: StudentID (integer) and StudentName (varchar, up to 100 characters).

- The StudentID must be the Primary Key.
- The StudentName cannot be left empty (must use a constraint).

```
CREATE TABLE StudentDetails(  
    StudentID INT PRIMARY KEY,  
    Student Name VARCHAR(20) NOT NULL  
);
```

Activity 2

Question: Create a table named LibraryBooks to manage library inventory. It needs three columns:

1. BookID (integer) - Set this as the Primary Key.
2. ISBN (varchar, 20 characters) - This number must be unique for every book.
3. Status (varchar, 20 characters) - If no status is provided when adding a book, it should automatically be set to 'Available'.

```
CREATE TABLE LibraryBooks(  
    BookID INT PRIMARY KEY,  
    ISBN VARCHAR(13) UNIQUE,  
    Status VARCHAR(20) DEFAULT 'Available'  
);
```

Activity 3

Question: Create a table named Employees. Include the following columns: EmpID (integer, Primary Key), Name (varchar, 100 characters, not null), and Age (integer).

- Add a constraint to ensure that the Age entered into the table is always 18 or older.

```
CREATE TABLE Employee(  
    EMPID INT PRIMARY KEY,  
    Name VARCHAR(20) NOT NULL,  
    Age INT CHECK (Age>=18)  
);
```

Activity 4

Question: You need to create a relationship between two tables to understand how Foreign Keys work.

1. First, create a Departments table with DeptID (integer, Primary Key) and DeptName (varchar, 50 characters, not null).
2. Second, create a Staff table with StaffID (integer, Primary Key), StaffName (varchar, 100 characters, not null), and DepartmentID (integer).
- Make the DepartmentID in the Staff table a Foreign Key that references the DeptID in the Departments table.

```
mysql> CREATE TABLE Staff(  
-> StaffID INT PRIMARY KEY,  
-> StaffName VARCHAR(100) NOT NULL,  
-> DepartmentID INT,  
-> FOREIGN KEY(DepartmentID) REFERENCE Department(DepID) ON DELETE CASCADE  
-> );
```

```
mysql> CREATE TABLE Staff(  
-> StaffID INT PRIMARY KEY,  
-> StaffName VARCHAR(100) NOT NULL,  
-> DepartmentID INT,  
-> FOREIGN KEY(DepartmentID) REFERENCES Department(DepID)  
-> );  
Query OK, 0 rows affected (0.05 sec)  
  
mysql> DESC Staff;  
+-----+-----+-----+-----+-----+-----+  
| Field          | Type          | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| StaffID        | int           | NO   | PRI | NULL    |       |  
| StaffName      | varchar(100)  | NO   |     | NULL    |       |  
| DepartmentID   | int           | YES  | MUL | NULL    |       |  
+-----+-----+-----+-----+-----+-----+
```

Activity 5

Write the SQL commands to modify this already existing table to:

1. Make CustomerID the Primary Key.

```
ALTER TABLE CustomerInfo  
ADD PRIMARY KEY(CustomerID);
```

```
mysql> CREATE TABLE Products(  
-> ProductID INT PRIMARY KEY,  
-> ProductName VARCHAR(50) NOT NULL,  
-> Price DECIMAL(10,2) CHECK (Price > 0),  
-> CategoryID INT,  
-> FOREIGN KEY(CategoryID) REFERENCES Categories(CategoryID)
```

```

-> );
INSERT INTO Categories(CategoryID,CategoryName)
-> VALUES
-> (1,'Electronics'),
-> (2,'Clothing');
INSERT INTO Products(ProductID,ProductName,Price,CategoryID)
-> VALUES
-> (101,'SmartPhone',45000,1);
Query OK, 1 row affected (0.01 sec)

```

Activity 2: School Course Enrollment System

Question:

1. Create a database named **SchoolDB**.
2. Create a table named **Students** with **StudentID** (Int, Primary Key), **FullName** (Varchar 100, Not Null), and **Email** (Varchar 100, Unique).
3. Create a table named **Courses** with **CourseID** (Int, Primary Key), **CourseName** (Varchar 100, Not Null), and **Credits** (Int, Default value is 3).
4. Create a junction table named **Enrollments** to link students and courses. It needs:
 - o **EnrollmentID** (Int, Primary Key)
 - o **StudentID** (Int, Foreign Key referencing **Students**)
 - o **CourseID** (Int, Foreign Key referencing **Courses**)
 - o **EnrollmentDate** (Date, Not Null)
5. Insert 2 students, 2 courses, and then enroll those students in the courses by inserting 3 records into the **Enrollments** table.

```

mysql> CREATE DATABASE SchoolDB;
Query OK, 1 row affected (0.02 sec)

mysql> USE SchoolDB;
Database changed
mysql> CREATE TABLE Student(
-> StudentID INT PRIMARY KEY,
-> FullName VARCHAR(100) NOT NULL,
-> Email VARCHAR(100) UNIQUE
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE Courses(
-> CourseID INT PRIMARY KEY,
-> CourseName VARCHAR(100) NOT NULL,
-> Credits INT DEFAULT 3
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE Enrollments(
-> EnrollmentID INT PRIMARY KEY,
-> StudentID INT,
-> CourseID INT,

```

```
-> EnrollmentDate DATE NOT NULL,  
-> FOREIGN KEY(StudentID) REFERENCES Student(StudentID),  
-> FOREIGN KEY(CourseID) REFERENCES Courses(CourseID)  
-> );
```

Query OK, 0 rows affected (0.05 sec)

Activity 3: Hospital Appointment System

Question:

1. Create a database named **HospitalDB**.
2. Create a **Doctors** table: **DoctorID** (Int, Primary Key), **DocName** (Varchar 100, Not Null), **Specialization** (Varchar 50, Not Null).
3. Create a **Patients** table: **PatientID** (Int, Primary Key), **PatientName** (Varchar 100, Not Null), **Age** (Int, Check age is ≥ 0), **ContactNo** (Varchar 15, Unique).
4. Create an **Appointments** table:
 - o **AppID** (Int, Primary Key)
 - o **PatientID** (Int, Foreign Key to **Patients**)
 - o **DoctorID** (Int, Foreign Key to **Doctors**)
 - o **AppStatus** (Varchar 20, Default 'Scheduled')
5. Insert 2 Doctors, 3 Patients, and 3 Appointments.

```
mysql> CREATE DATABASE HospitalDB;  
Query OK, 1 row affected (0.04 sec)
```

```
mysql> USE HospitalDB;  
Database changed  
mysql> CREATE TABLE Doctors(  
-> DoctorID INT PRIMARY KEY,  
-> DocName VARCHAR(100) NOT NULL,  
-> Specialization VARCHAR(50) NOT NULL  
-> );  
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE Patients (  
-> PatientID INT PRIMARY KEY,  
-> PatientName VARCHAR(100) NOT NULL,  
-> Age INT CHECK (Age >= 0),  
-> ContactNo VARCHAR(15) UNIQUE  
-> );  
Query OK, 0 rows affected (0.05 sec)
```

```
mysql>  
mysql> -- Step 4: Create Appointments table connecting Doctors and Patients  
mysql> CREATE TABLE Appointments (  
-> AppID INT PRIMARY KEY,  
-> PatientID INT,  
-> DoctorID INT,
```

```
-> AppStatus VARCHAR(20) DEFAULT 'Scheduled',
-> FOREIGN KEY (PatientID) REFERENCES Patients(PatientID),
-> FOREIGN KEY (DoctorID) REFERENCES Doctors(DoctorID)
-> );
```

Query OK, 0 rows affected (0.03 sec)

mysql> -- Step 5: Insert Data

mysql> -- Insert Doctors

mysql> INSERT INTO Doctors (DoctorID, DocName, Specialization) VALUES

```
-> (1, 'Dr. Samantha', 'Cardiologist'),
```

```
-> (2, 'Dr. Ruwan', 'Dermatologist');
```

Query OK, 2 rows affected (0.00 sec)

Records: 2 Duplicates: 0 Warnings: 0

mysql> -- Insert Patients

mysql> INSERT INTO Patients (PatientID, PatientName, Age, ContactNo) VALUES

```
-> (10, 'Sunil Jayawardena', 45, '0771234567'),
```

```
-> (11, 'Amala Fernando', 30, '0719876543'),
```

```
-> (12, 'Kasun Bandara', 22, '0701112233');
```

Query OK, 3 rows affected (0.00 sec)

Records: 3 Duplicates: 0 Warnings: 0

mysql> -- Insert Appointments

mysql> INSERT INTO Appointments (AppID, PatientID, DoctorID) VALUES

```
-> (1001, 10, 1),
```

```
-> (1002, 11, 2);
```

Query OK, 2 rows affected (0.00 sec)

Records: 2 Duplicates: 0 Warnings: 0

mysql> -- Insert an Appointment with a specific status (Overriding the default)

mysql> INSERT INTO Appointments (AppID, PatientID, DoctorID, AppStatus) VALUES

```
-> (1003, 12, 1, 'Completed');
```

Query OK, 1 row affected (0.00 sec)